Muhammad Afan Tahir

West Haven, CT | afantahir.work@gmail.com | 203-410-7152 | linkedin.com/in/afantahir | afantahir.github.io

Full-time from May 2027 (M.S. expected May 2027) | Driver's license & own vehicle | Willing to travel and relocate

SUMMARY

Civil engineer completing an M.S. at the University of New Haven (May 2027, GPA 3.78) with hands-on construction and inspection experience. Supervised crack repair and structural rehabilitation on 5+ occupied sites, including a major shopping mall and a pedestrian bridge, with **every RFI accepted** by the consultant; inspected an in-service Connecticut bridge and prepared the report to **FHWA NBIS and CTDOT standards**; holds **six FHWA-NHI inspection certificates, CMIT, OSHA 30 and LEED GA**, with the FE scheduled for December 2026. Teaching Assistant to the Department Chair and President of the SEI Graduate Chapter. Seeking **inspection, field engineering, project engineering or construction management** roles with DOTs, consultants and contractors.

EDUCATION

University of New Haven — M.S. Civil Engineering | Provost Assistantship (2026–2027) May 2027 | GPA 3.78/4.0
Bridge Inspection; Construction Inspection & Monitoring; Building Inspection; Steel Design; Foundation Design
MAHSA University, Malaysia — B.Eng. Civil Engineering (Hons) | 8× Dean's Award; Gold Award, Thesis Feb 2025 | GPA 3.91/4.0

CERTIFICATIONS

- NHI 130101: Safety Inspection of In-Service Bridges 2025
- NHI 130128: Bridge Construction Inspection 2025
- NHI 130087A: Inspection of Ancillary Structures 2025
- NHI 130111: Nondestructive Evaluation (NDE) 2026
- NHI 134108: Highway Plan Reading Series (8 modules) 2026
- CMIT – Construction Manager in Training (CMAA) 2025
- OSHA 30-Hour Construction Safety & Health 2026
- LEED Green Associate (USGBC) – scored 195/200 2026
- NHI 130124: Tunnel Safety Inspection 2026
- FE Civil Exam (scheduled) Dec 2026
- LinkedIn Learning (PMI PDUs): CM Foundations; Scheduling & Procurement; Primavera P6 | Drone Scanning Workshop 2024–2026

SUPERVISED ACADEMIC BRIDGE INSPECTION

Bridge No. 03017 Routine Inspection — Assistant Team Leader | Dr. Byungik Chang, PE New Haven, CT | Oct–Dec 2025

- **Inspected** an in-service bridge with the team: reviewed prior CTDOT records, walked abutments and superstructure, logged defects with photos.
- **Drafted** plan and elevation sketches in Bluebeam Revu and **mapped every observed defect** (cracks, hollow areas, patches, efflorescence) to abutment and girder line.
- **Delivered** the inspection report to **FHWA NBIS and CTDOT standards**: condition ratings, defect inventory and repair recommendations.

PROFESSIONAL EXPERIENCE

University of New Haven, Civil & Environmental Eng. — Graduate Teaching Assistant West Haven, CT | Aug 2026–Present

- **Selected** by the Department Chair (Dr. Byungik Chang, PE) for the Provost Assistantship; **run** AutoCAD (CIVL 1109) and Materials Lab (CIVL 3323) sessions, lab QA/QC and departmental projects.

cmk LEED — LEED Project Trainee | Cynthia Kaplan, LEED AP BD+C South Windsor, CT | Mar 2026–Present

- **Earned** LEED Green Associate at 195/200, then **completed** 15 hrs of continuing education on live projects; **contribute** to credit documentation and Gold/Platinum certification strategy.

Structural Healthcare Research Associates — Project Engineer (part-time) Kuala Lumpur, Malaysia | May 2024–Feb 2025

- **Jalan Ampang Pedestrian Bridge:** executed the dimensional survey, **produced** as-built AutoCAD drawings, mapped defects and drafted the remedial works plan for the structural integrity appraisal.

RNC Concrete Repair Specialist — Assistant Project Engineer Intern Selangor, Malaysia | Aug 2023–Apr 2024

- **Supervised** 5+ concrete rehabilitation projects end-to-end: field inspection, crack mapping, before/after defect registers and crew supervision.
- **Pavilion Mall:** directed crack rectification across four levels of an occupied mall; **every RFI accepted** by the consultant at monthly claims.
- **Certified** repair quantities against approved plans and specifications; **compiled** daily site records, RFIs and BOQs that passed consultant audit.

LEADERSHIP

SEI Graduate Student Chapter, University of New Haven — President (elected) Aug 2026–Present

- **Lead** a 5-member executive board and the chapter's annual programme, budget and industry outreach; **launched** weekly FE exam prep, speaker seminars, resume review nights and construction site visits with local firms.

RESEARCH & HONORS

- **Published** two journal papers on green concrete blocks from agricultural and plastic waste (MAHSA Int. J. Eng. & Tech., 2024); **Gold Award** ESTE'25, **Best Presenter** Mi-SITE'24, **Best Poster** ESTE'24.

SKILLS & AFFILIATIONS

Field & QA/QC: NBI ratings, crack mapping, photo logs, plan/spec compliance, daily reports, quantities, RFIs, submittals, BOQs
Software: AutoCAD, Civil 3D, Revit, Navisworks, Bluebeam, Tekla SD, MS Project, Primavera P6, SAFE, Excel
Memberships: ASCE, SEI, ACI, CMAA | **Honors:** First Prize, Bridge Competition

REFERENCE

Byungik Chang, Ph.D., PE, MBA, F.ASCE, F.SEI — Professor & Chair, Civil & Environmental Engineering, University of New Haven
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